



Vector Control, Pest Management, Resistance, Repellents

Molecular detection of insecticide resistance markers in *Aedes aegypti* (Diptera: Culicidae) populations from Haiti

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Aedes aegypti (Linnaeus, 1762) mosquitoes are the primary vectors of several arboviruses of major public health importance, including dengue virus, Zika virus, chikungunya virus, and yellow fever virus. In Haiti, *Ae. aegypti* mosquitoes are highly abundant and linked to the transmission and spread of these pathogens. As the use of pesticides for control of mosquitoes increases, there is heightened selection pressure for insecticide-resistant (IR) mosquitoes, mitigating the efficacy of pesticides and leading to an increased risk of continual pathogen transmission. To this end, active surveillance of resistance status can inform more effective operational control strategies. In this study, we screened 421 individual *Ae. aegypti* mosquitoes collected from the Ouest department of Haiti between 2018 and 2019 for the presence of IR mutations S989P, F1534C, V1016I, and V1016G. We observed IR-conferring alleles in all study sites, including a high prevalence of the homozygous resistance variant of F1534C across time, an increased prevalence of the homozygous resistance variant V1016I across time (an estimated 12.27% higher odds across time), and the absence of resistance-associated alleles for S989P. Our results indicate that pyrethroid resistance is prevalent and increasing in *Ae. aegypti* populations within this region of Haiti.

Keywords: insecticide resistance, kdr, *Aedes aegypti*, Haiti

Introduction

Aedes aegypti (Linnaeus, 1762) mosquitoes are the primary vectors of chikungunya virus (CHIKV), yellow fever virus (YFV), Zika virus (ZIKV), and dengue virus (DENV). The incidence of these diseases has increased dramatically across the globe within the past 30 years, with current estimates of the annual number of cases of these mosquito-borne viruses ranging from 84,000 (YFV) to over 130 million (DENV) (Franklin et al. 2019). Although these pathogens generally have low attributable mortality rates amongst populations with access to effective medical treatment, the global economic impact

related to treatment and losses in productivity is significant, with the economic burdens of DENV and ZIKV both estimated at 8.9 billion USD (Laserna et al. 2018, Chilakam et al. 2023). Due to a lack of vaccines, the most effective method to prevent transmission of these arboviruses is mosquito control (Severson and Behura 2016).

Within Haiti, numerous mosquito-borne arboviruses have been reported, including CHIKV, DENV, YFV, and ZIKV. According to the Pan American Health Organization, the annual number of dengue cases in Haiti ranged from 90 to 6,298 between 2018 and 2022, while there have been no

reported cases of chikungunya or Zika since 2019. However, a recent study suggests a continued low transmission rate of ZIKV and CHIKV (Louis et al. 2025). DENV is of particular concern, given that seroprevalence rates of anti-DENV antibodies are as high as 72% in Haiti, and reinfection with different serotypes of DENV can elicit more severe and potentially fatal symptoms, such as dengue shock syndrome (Weppelmann et al. 2017). The endemic nature of DENV and the continued persistence of CHIKV and ZIKV within Haiti underscore the need for concerted disease prevention measures, including strategies that target mosquito populations.

Treated bed nets and pyrethroid aerosol spray cans are two widely adopted *Ae. aegypti* control strategies in Haiti (Estep et al. 2024). Mosquito control efforts are challenged by the emergence of insecticide resistance, which is exacerbated by significant resource limitation that negatively impacts management and surveillance efforts throughout the country. (Estep et al. 2024 Jun 1). Geographical and socioeconomic vulnerabilities to environmental hazards further hamper Haiti's ability to establish effective and lasting mosquito management programs (Diouf et al. 2024). As such, it could be expected that pyrethroid resistance may be increasing over time within Haiti, as has been demonstrated in other studies (Baltzegar et al. 2021). Pyrethroid resistance, or knockdown resistance (*kdr*), mechanisms in *Aedes* mosquitoes are relatively well-studied. Pyrethroids bind to insect voltage-sensitive sodium channels (VSSCs) and lead to desensitization, resulting in sustained paralysis and death (Du et al. 2016). Twelve single amino acid point mutations have been associated with resistance to pyrethroids in *Ae. aegypti*, with 5 having been functionally confirmed thus far: S989P (IIL5–6), I1011M, V1016I/G (IIS6), F1534C (IIIS6), and V410L (IS6) (Saavedra-Rodriguez et al. 2018, Fan et al. 2020). In this study, we focused on screening for single-nucleotide polymorphisms (SNPs) in 3 functionally confirmed resistance loci, S989, F1534, and V1016 within *Ae. aegypti* mosquitoes collected from the Ouest Department of Haiti.

Materials and Methods

Collection Sites and Mosquito Collections

Mosquitoes were collected at 4 study sites within Haiti's Ouest Department (Fig. 1). Mosquitoes were collected using 3

mosquito trap types (Biogents Sentinel Traps, Centers for Disease Control and Prevention [CDC] Gravid Traps, and CDC Light Traps). Collections were performed at least once at each site between September 2018 and August 2019.

Mosquito Sample Processing and Nucleic Acid Extraction

Genomic DNA (gDNA) was extracted from individual, whole *Ae. aegypti* mosquitoes using Qiagen's DNeasy Blood & Tissue kit (Qiagen, Germantown, MD) according to their protocol "Purification of Total DNA from Insects using the DNeasy Blood & Tissue kit," with the following modifications: each mosquito was washed 3 times in 1X PBS and then mechanically homogenized in 180 µl of 1X PBS for 3 min at 1,200 rpm using a BeadBlaster 96 Ball Mill Homogenizer (Benchmark Scientific, Sayreville, New Jersey). DNA concentrations were assessed using a Qubit fluorometer (ThermoFisher Scientific, Waltham, Massachusetts).

qPCR Detection of *Kdr* Mutations

SNP genotyping for F1534C, V1016G, V1016I, and S989P was performed on each mosquito gDNA sample using TaqMan assays developed by Sene et al. (2021) with modifications as listed. F1534C and S989P were duplexed, while V1016I/G was triplexed. Primer and probe sequences are listed in Table 1. We redesigned 989_Probe-mt due to poor performance during assay validation and found 989P_Probe-mtB to be an adequate alternative. TaqMan Fast Advanced Mastermix (ThermoFisher Scientific, Waltham, Massachusetts) was used, and all reactions were run in 20 µl reaction volumes on a QuantStudio 6 Flex qPCR instrument (Applied Biosystems, Waltham, Massachusetts). Synthetic gBlocks (Table 2) were used as positive controls on each plate to verify proper assay performance as in Sene et al. (2021). Non-template controls were also included on each plate. The thermocycler conditions were 50°C for 2 min, 95°C for 2 min, and 40 cycles of 95°C for 3 s and 60°C for 30 s.

Sequencing for qPCR Validation

The qPCR results were confirmed in 105 mosquito samples by amplicon sequencing of the *Ae. aegypti* VSSC domain II (for S989P and V1016I/G mutations) and domain III (for the

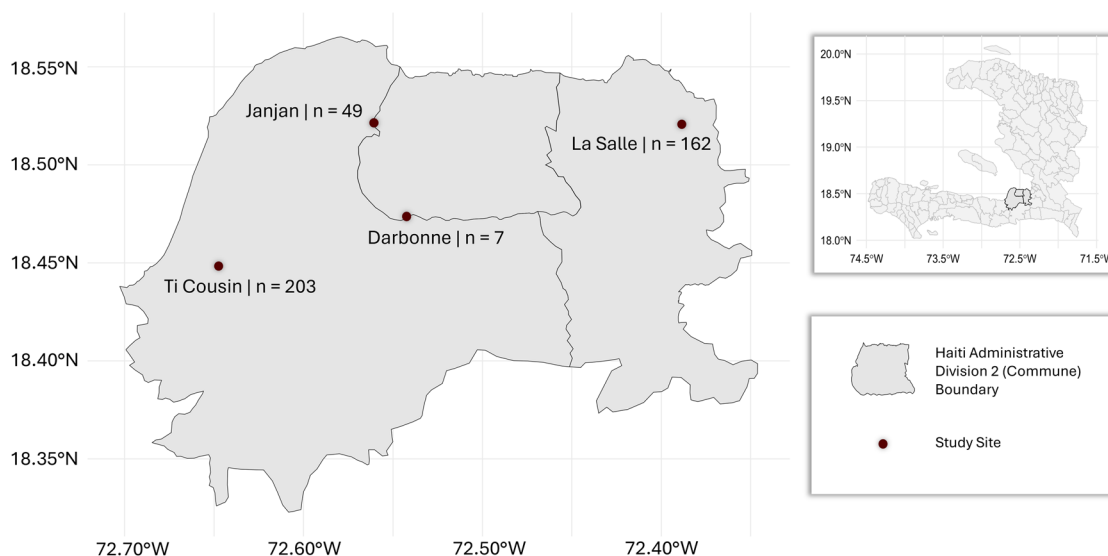


Fig. 1. Geographic location of sampling sites along with number of samples collected and analyzed per location.

Table 1. Primer and probe sequences used for initial qPCR-based *kdr* mutation detection assays, sequencing-based *kdr* mutation validation, and sequencing-based arginine kinase gene mosquito species validation

Assay	Reagent	Oligo_Name	Sequence (5' - 3')	Length (bp)	Assay description	5'mod	3'mod	Optimized Concentration (nM)
qPCR	Primer	kdr_F1534C-F	CGAGACCAACATCTA-CATGTACCT	24	A. aegypti kdr_Vgsc-F1534C F			400
	Primer	kdr_F1534C-R	GTCGATGATGA-CACCGATGA	20	A. aegypti kdr_Vgsc-F1534C R			500
	Probe	F1534_P-wt	AACGACCCGAAGATGA	16	A. aegypti kdr_Vgsc-F1534 - wild type	NED	MGB	800
	Probe	1534C_P-mt	ACGACCCGCAGATGA	15	A. aegypti kdr_Vgsc-C1534 - mutant	FAM	MGB	200
	Primer	kdr_S989P-F	CATGATCGT-GTTCCGGGTATT	21	A. aegypti kdr_Vgsc-S989P F			450
	Primer	kdr_S989P-R	CACGTCACCCA-CAAGCATACA	21	A. aegypti kdr_Vgsc-S989P R			900
	Probe	S989_P-wt	ATCGAATCCATGTGGGA	17	A. aegypti kdr_Vgsc-S989 - wild type	NED	MGB	750
	Probe	989P_Probe-mt	AGTGGATCGAACCCA	15	A. aegypti kdr_Vgsc-P989 - mutant	FAM	MGB	NA
	Probe	989P_Probe-mt B	ATCGAACCCATGTGGG	16	A. aegypti kdr_Vgsc-P989 - mutant	FAM	MGB	100
	Primer	kdr_V1016- F	ACCGA-CAAATTGTTTCCAC	20	A. aegypti kdr_Vgsc-V1016G F			400
	Primer	kdr_V1016- R	GGACAAAAG-CAAGGCTAAGAA	21	A. aegypti kdr_Vgsc-V1016G R			300
	Probe	V1016_P-wt	CACAGGTACTTAACCT	16	A. aegypti kdr_Vgsc-V1016 wild type	NED	MGB	500
	Probe	1016G_P-mt1	CACAGGGACTTAACC	15	A. aegypti kdr_Vgsc-1016G—mutant	FAM	MGB	100
	Probe	1016I_P-mt2	CGCACAGATACTTAA	15	A. aegypti kdr_Vgsc-1016I—mutant	VIC	MGB	300
amp-seq	Primer	DII 1016_989 F	GGTG-GAACTTCACCGACTT	19	A. aegypti domain II 989 to 1016 F			
	Primer	DII 1016_989 R	GGACGCAATCT-GGCTTGTTA	20	A. aegypti domain II 989 to 1016 R			
	Primer	DIII F1534C F	GCTGTGCGAC-GAGATCATT	19	A. aegypti domain III 1534 F			
	Primer	DIII F1534C R	GTTGAAC-CCGATGAACAACA	20	A. aegypti domain III 1534 R			
	Primer	akF	GCTTCAAGAAGAC-CGACAAGCAC	23	arginine kinase			
	Primer	akF2	AAGACCTTCCTG-GTCTGGTGC	21				
	Primer	akR	ACCCWKCTGCATS-GAGATGATG	22				
	Primer	akR2	GCCATCGTA-CATCTCCTTGACG	22				

F1534C mutation) genes. PCRs were carried out in 50 µl reaction volumes using Q5 2X High-Fidelity Master Mix (New England Biolabs, Ipswich, Massachusetts), 10 µM of forward and reverse primers, 10 µl of nuclease-free water, and 5 µl of template DNA. The thermal cycler settings were 98 °C for 30 s, then 35 cycles of 98 °C for 5 s, 65 °C for 20 s, 72 °C for 30 s, followed by 72 °C final extension for 2 min. Additionally, the arginine kinase (AK) gene was amplified and sequenced for species confirmation using primers from [Reidenbach et al. \(2009\)](#). AK PCRs were carried out in 50 µl reactions using Q5 2X High-Fidelity Master Mix, 10 µM of forward and reverse primers, 12 µl of nuclease-free water, and 8 µl of template DNA. All primers are listed in [Table 1](#). The AK thermal cycler settings were 95 °C for 3 min, then 35 cycles of 94 °C for 10 s, 60 °C for 40 s, 72 °C for 30 s, followed by 72 °C final extension

for 2 min. *Kdr* and AK amplicons for each sample were library prepped using the Illumina DNA Prep kit (Illumina, San Diego, California) and run on either a MiSeq V3 600-cycle cartridge or a NextSeq P2 600-cycle XLEAP-SBS cartridge on a MiSeq or NextSeq 1000, respectively.

Bioinformatics

The resulting FASTQ files were quality checked using FastQC (version 0.11.9, Babraham Bioinformatics, Cambridge, United Kingdom) to assess the quality scores, adapter content, and other quality metrics ([Andrews 2010](#)). Amplicon reads were trimmed using the bbdut.sh with the following parameters: ref={PATH_2_bmap}/bmap/resources/adapters.fa, ktrim=r, k=23, mink=11, hdist=1, tpe, tbo, qtrim=r, trimq=28, minlen=100,maq=20, filterpolyg=20, filterpolyc=20, trimpolya=20, trimpolygright=20,

Table 2. Synthetic gBlock sequences used as positive controls for qPCR-based *kdr* mutation detection assays (Sene et al. 2021)

Name	Sequence	Notes
GB1—wild type sequences	TTCGCGAGACCAACATCTACATGTACCTCTACTTTGTGTTCTTCATCATCTTCG-GGTCGTTCTTCACGCTGAATCTGTTTCATCGGTGTCATCATCGACAATTCAT-GATCGTGTTCGGGTATTGTGCGGCGAGTGGATCGAATCCATGTGGGATTG-TATGCTTGTGGGTGACGTGTCTACCGACAAATTGTTTCCCACTCGCACAG-GTACTTAACCTTTTCTTAGCCTTGCTTTTGTC	For wt genotypes F1534, S989, & V1016
GB2—mutant sequences (1016 mt2)	TTCGCGAGACCAACATCTACATGTACCTCTACTTTGTGTTCTTCATCATCTGCG-GGTCGTTCTTCACGCTGAATCTGTTTCATCGGTGTCATCATCGACAATTCAT-GATCGTGTTCGGGTATTGTGCGGCGAGTGGATCGAATCCATGTGGGATTG-TATGCTTGTGGGTGACGTGTCTACCGACAAATTGTTTCCCACTCGCACAGA-TACTTAACCTTTTCTTAGCCTTGCTTTTGTC	For mutants 1534C, 989P, & 1016I
GB3—mutant sequence 1 for 1016	TTCGCGAGACCAACATCTACATGTACCTCTACTTTGTGTTCTTCATCATCTTCG-GGTCGTTCTTCACGCTGAATCTGTTTCATCGGTGTCATCATCGACAATTCAT-GATCGTGTTCGGGTATTGTGCGGCGAGTGGATCGAATCCATGTGGGATTG-TATGCTTGTGGGTGACGTGTCTACCGACAAATTGTTTCCCACTCGCACAG-GGACTTAACCTTTTCTTAGCCTTGCTTTTGTC	For mutant 1016G

entropy=0.5, entropywindow=50, entropyk=5, entropytrim=rl (Bushnell 2014). *De novo* assembly of the trimmed reads was then carried out using the program minia with the following parameters: kmer-size=51, abundance-min=250 (Chikhi and Rizk 2013). Additionally, reads from each arginine kinase and voltage sensitive sodium channel amplicon set were aligned to an *Ae. aegypti* (GCF_002204515.2) reference genome using bowtie2 (Langmead and Salzberg 2012, Matthews et al. 2018). Read group details and duplicates were marked using the AddOrReplaceReadGroups and MarkDuplicates software within the Genome Analysis Toolkit (GATK) (Auwerda 2020). Reads covering the targeted regions of the voltage sensitive sodium channel and AK genes were assessed with samtools coverage and manually using the Integrative Genomics Viewer (IGV) (Li et al. 2009, Robinson et al. 2011). SNPs associated with known insecticide resistance alleles (S989P, V1016G/I, and F1534C) were assessed manually in IGV (Wang et al. 2023). In addition, the SNPs were called using the bcftools (v1.11) mpileup and call tools with a relaxed prior of 0.1 in the multiallelic-caller mode (Li et al. 2009). Consensus sequences for the AK regions were generated using bcftools consensus and aligned against reference mosquito AK regions using the muscle aligner in MEGA11 (Edgar 2004). MEGA11 was then used to generate a phylogenetic tree (Supplementary Fig. S1) using the GTR and BioNJ parameters (Tamura et al. 2021). *Ae. aegypti* AK sequences were uploaded to the Sequence Read Archive (SRA) with accessions SRR33623067-SRR33623080, SRR33623082-SRR33623108, SRR33623110-SRR33623118, SRR33623120-SRR33623130, and SRR33623132-SRR33623141. *Ae. aegypti kdr* sequences were uploaded to the SRA with accessions SRR33623067-SRR33623143. All uploaded sequences are under BioProject PRJNA1265117.

Data Analysis

Data analysis and figure creation were performed using RStudio version 4.3.2. Mapping and data visualizations were created using the R packages *readr*, *stringr*, *lubridate*, *scales*, *sf*, *dplyr* and *tidyr* (Grolemund and Wickham 2011, Pebesma and Bivand 2023, Wickham 2023, Wickham et al. 2023a, 2023b, 2023c, 2024). All code and data for the statistical tests is available at https://github.com/IanPsheaSmith/Haiti_Aeae_IR or upon request to the authors.

Fisher's exact test was used to assess whether there were statistically significant differences in rates of susceptible and resistant alleles across the 4 study sites. Fisher's exact test provides a more conservative estimate of difference across contingency tables and is particularly useful when sample sizes are small (Fisher 1922). To assess differences across time, we employed the same test but used a Monte Carlo simulation approach to obtain the *P*-value (the number of simulated random tables was set at 1,000,000 to ensure a highly stable estimate). The alpha level for hypothesis testing was set at 5%. One sample at the La Salle site was omitted from the temporal analysis due to uncertainty about its collection date.

To test whether the proportion of mosquito samples that were homozygous for resistance alleles across time was increasing, we employed a logistic regression model with month (treated as a continuous, numeric variable) as the predictor and a binary comparator of homozygous resistant alleles to non-homozygous resistant alleles as the outcome variable (Hosmer et al. 2000). The logistic regression model took the form $\log(P/(1-P)) = \beta_0 + \beta_1 \text{Month}$, where *P* is the probability of observing the homozygous resistant genotype (R/R) and *Month* is treated as a continuous variable, increasing numerically across the study's timeframe.

Results

A total of 421 *Ae. aegypti* mosquitoes were captured and analyzed in this study. Tables 3 and 4 demonstrate the genotype frequencies, by location, and the genotypic qPCR-NGS concordance, respectively. Tables 5 and 6 demonstrate the allelic frequencies, by location, and allelic qPCR-NGS concordance. While 100% of samples were homozygous susceptible at the 989 locus, no samples were homozygous susceptible at the 1534 locus and only 2.6% of samples were homozygous susceptible at the 1016 locus. Over 95% of samples were homozygous resistant at the 1534 locus, while 72.7% of samples were homozygous resistant at the 1016 locus, with 24.7% of samples heterozygous at the 1016 locus. The occurrence of homozygous resistance at locus 1016 ranged from 70.0% to 87.8% by location, while homozygous resistance at locus 1534 occurred in 97% to 100% of samples by location. The concordance of the qPCR and NGS results at the location level ranged

Table 3. *Kdr* genotype frequencies by location

Location	S989P			V1016I			F1534C		
	RR	RS	SS	RR	RS	SS	RR	RS	SS
Darbonne (<i>n</i> =7)	-	-	7 (100%)	5 (71.4%)	2 (28.6%)	-	7 (100%)	-	-
Janjan (<i>n</i> =49)	-	-	49 (100%)	43 (87.8%)	5 (10.2%)	1 (2.0%)	48 (98.0%)	1 (2.0%)	-
La Salle (<i>n</i> =162)	-	-	162 (100%)	116 (71.6%)	43 (26.5%)	3 (1.8%)	158 (97.5%)	4 (2.5%)	-
Ti Cousin (<i>n</i> =203)	-	-	406 (100%)	142 (70.0%)	54 (26.6%)	7 (3.4%)	197 (97.0%)	6 (3.0%)	-
Total (<i>n</i> =421)	-	-	421 (100%)	306 (72.7%)	104 (24.7%)	11 (2.6%)	410 (97.4%)	11 (1.3%)	-
Fisher's exact test <i>P</i> -value	1			0.1804			1		

Percentages demonstrate the number of samples with specific genotypes relative to total number of samples tested for the specific location and/or total number of samples. Fisher's exact test was used to assess whether there were statistically significant differences in rates of susceptible and resistant alleles across the 4 study sites, with no statistically significant differences found at an alpha level of 5%.

Table 4. *Kdr* genotype qPCR-NGS concordance

Location	S989P			V1016I			F1534C		
	RR	RS	SS	RR	RS	SS	RR	RS	SS
Darbonne (<i>n</i> =7)	-	-	7 (100%)	4 (80.0%)	2 (100%)	-	7 (100%)	-	-
La Salle (<i>n</i> =44)	-	-	44 (100%)	34 (100%)	10 (100%)	-	43 (100%)	1 (100%)	-
Ti Cousin (<i>n</i> =54)	-	-	54 (100%)	38 (92.7%)	10 (100%)	-	52 (98.1%)	1 (100%)	-
Total (<i>n</i> =105)	-	-	105 (100%)	76 (95.0%)	22 (100%)	-	102 (99.0%)	2 (100%)	-
Fisher's exact test <i>P</i> -value	1			0.8869			1		

Values represent the number of samples whose NGS-determined genotype matched the qPCR-determined genotype. Percentages demonstrate the proportion of samples where the NGS- and qPCR-determined genotypes were concordant. Fisher's exact test was used to assess whether there were statistically significant differences in rates of susceptible and resistant alleles across the 4 study sites, with no statistically significant differences found at an alpha level of 5%.

Table 5. *Kdr* allele frequencies by location

Location	S989P		V1016I		F1534C	
	R	S	R	S	R	S
Darbonne (<i>n</i> =7)	-	14 (100%)	12 (85.7%)	2 (14.3%)	14 (100%)	-
Janjan (<i>n</i> =49)	-	98 (100%)	91 (92.8%)	7 (7.2%)	97 (99.0%)	1 (1.0%)
La Salle (<i>n</i> =162)	-	324 (100%)	275 (84.9%)	49 (15.1%)	320 (98.8%)	4 (1.2%)
Ti Cousin (<i>n</i> =203)	-	406 (100%)	338 (83.2%)	68 (16.8%)	400 (98.5%)	6 (1.5%)
Total (<i>n</i> =421)	-	842 (100%)	716 (85.0%)	126 (15.0%)	831 (98.7%)	11 (1.3%)
Fisher's exact test <i>P</i> -value	1		0.09191		1	

Percentages demonstrate the number of samples with specific allele frequency relative to total number of samples tested for the specific location and/or total number of samples. Fisher's exact test was used to assess whether there were statistically significant differences in rates of susceptible and resistant alleles across the 4 study sites, with no statistically significant differences found at an alpha level of 5%.

Table 6. *Kdr* allele qPCR-NGS concordance

Location	S989P		V1016I		F1534C	
	R	S	R	S	R	S
Darbonne (<i>n</i> =7)	-	14 (100%)	11 (91.7%)	2 (100%)	14 (100%)	-
La Salle (<i>n</i> =44)	-	88 (100%)	78 (100%)	10 (100%)	87 (100%)	1 (100%)
Ti Cousin (<i>n</i> =54)	-	108 (100%)	90 (97.8%)	16 (100%)	103 (96.3%)	1 (100%)
Total (<i>n</i> =105)	-	210 (100%)	180 (98.9%)	28 (100%)	206 (99.04%)	2 (100%)
Fisher's exact test <i>P</i> -value		1	0.844		1	

The value represents the number of samples whose NGS-determined genotype matched the qPCR-determined genotype. Percentages demonstrate the proportion of samples where the NGS- and qPCR-determined alleles were concordant. Fisher's exact test was used to assess whether there were statistically significant differences in rates of susceptible and resistant alleles across the 4 study sites, with no statistically significant differences found at an alpha level of 5%.

from 80% to 100% for genotype and 91.7% to 100% for alleles.

Table 7 details the distribution of samples by *kdr* classification group. Overall, 72.4% of samples were homozygous resistant at both 1016 and 1534 (IICC, ranging from 70.0% to

85.7% by location), while no samples were homozygous susceptible at both 1016 and 1534 (VVFF). Of the remaining samples, 23% were heterozygous at 1016 but homozygous resistant at 1534 (ranging from 10.2% to 28.5% by location). All other *kdr* classification groups accounted for less than 5%

Table 7. Distribution of results by *kdr* classification group

Location	IICC	VICC	VIFC	VVCC	VVFC	VVFF
Darbonne (<i>n</i> =7)	5 (71.4%)	2 (28.5%)	-	-	-	-
Janjan (<i>n</i> =49)	42 (85.7%)	5 (10.2%)	-	1 (2.0%)	-	-
La Salle (<i>n</i> =162)	116 (71.6%)	39 (24.1%)	4 (2.5%)	3 (1.8%)	-	-
Ti Cousin (<i>n</i> =203)	142 (70.0%)	51 (25.1%)	3 (1.5%)	4 (2.0%)	3 (1.5%)	-
Total (<i>n</i> =421)	305 (72.4%)	97 (23.0%)	7 (1.7%)	8 (1.9%)	3 (0.7%)	-

IICC: homozygous resistant at both 1016 and 1534. VICC: heterozygous at 1016 but homozygous resistant at 1534. VIFC: heterozygous at both 1016 and 1534. VVCC: homozygous susceptible at 1016 and homozygous resistant at 1534. VVFC: homozygous susceptible at 1016 and heterozygous at 1534. VVFF: homozygous susceptible at both 1016 and 1534.

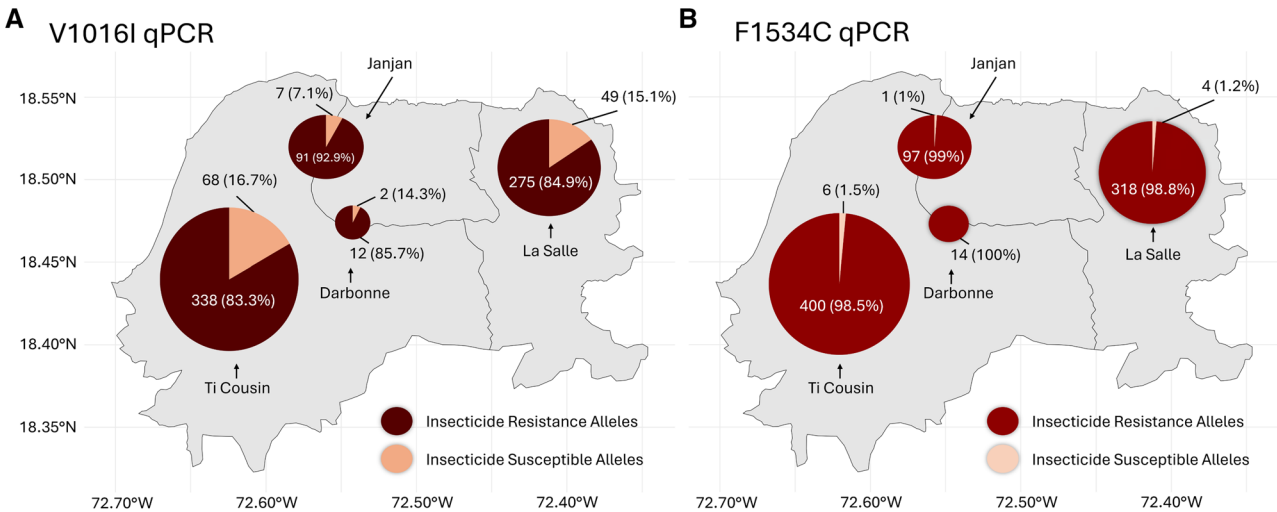


Fig. 2. Allele frequencies by study site based on qPCR. A shows counts and percentages for gene V1016I and B shows counts and percentages for F1534C. Results for S989P are not shown as all sites had a frequency of 100% susceptible. The size of the pie charts indicates the quantity of samples collected, with sites Ti Cousin (Southwest) and La Salle (Northeast) having the greatest quantity of samples. The proportion of resistance alleles is not significantly different across study sites based on Fisher's exact test, which holds true for all 3 alleles that were examined.

of samples. As a procedural control, 9 mosquitoes belonging to the Orlando strain of *Ae. aegypti* were analyzed from the Navy Entomology Center of Excellence's insectary. As expected in a naïve population, no SNPs were detected for any of the 9 at the 989, 1016, and 1534 loci, either by qPCR or NGS.

Resistance overTime (All Sites, Based on qPCR Data)

Geographically, the ratio of alleles associated with resistance was consistent across all 4 sites, with all 4 sites having full susceptibility per allele S989P, around 15% susceptibility per allele V1016I and around 1% susceptibility per allele F1534C (Tables 3 and 4; Fig. 2). The lack of geographic variation was confirmed using Fisher's exact test, which showed no statistically significant differences across the 4 sites at an alpha level of 5%. We describe this same lack of geographic variation when comparing the proportions of homozygosity across sites (S989P $P=1$, V1016I $P=0.06084$, F1534C $P=1$), and this was also seen for the NGS concordance results, which did not differ significantly across the 3 sites from which samples were tested.

Across time, we elucidate no significant differences in the presence of resistance for genes S989P and F1534C (Fig. 3, P -values=1 and 0.190, respectively, based on Fisher's exact test). We did however find significant differences in the monthly proportions for allele V1016I (P -value=0.0002). Moreover, our logistic regression model for V1016I estimated a 12.27%

increase per month in the odds of detecting homozygous resistant (R/R) genotypes (95% CI: 5.4% to 21.3%, $P=0.0006$) over the study period, and the model outperformed a null model reflected by a residual deviance of 23.34 on 10 degrees of freedom in comparison to the null model's deviance of 35.125 on 11 degrees of freedom. Given the frequencies of these 3 resistance loci, the change in frequency in the absence of gene flow or de novo mutation is only expected to occur for V1016I due to its intermediate occurrence, which is corroborated by the observed upward trend exhibited in this study.

Discussion

This study demonstrates high frequencies of genetic markers for pyrethroid resistance in *Ae. aegypti* populations within this region of the Ouest Department of Haiti. While no resistant allele was detected at the 989 locus, high frequencies of homozygous resistance genotypes were observed for the 1016 (>70%) and 1534 (97%) loci. Homozygous susceptible genotypes were found in less than 3% of samples at locus 1016, while no homozygous susceptible genotypes were observed at the 1534 locus. Concerning allelic frequencies, overall susceptible allelic frequencies were 15% and 1.3% at the 1016 and 1534 loci, respectively. Considering all locations, 72.4% of samples had dilocus *kdr* mutations, with no samples being homozygous susceptible at both loci. Finally, a 12.27%

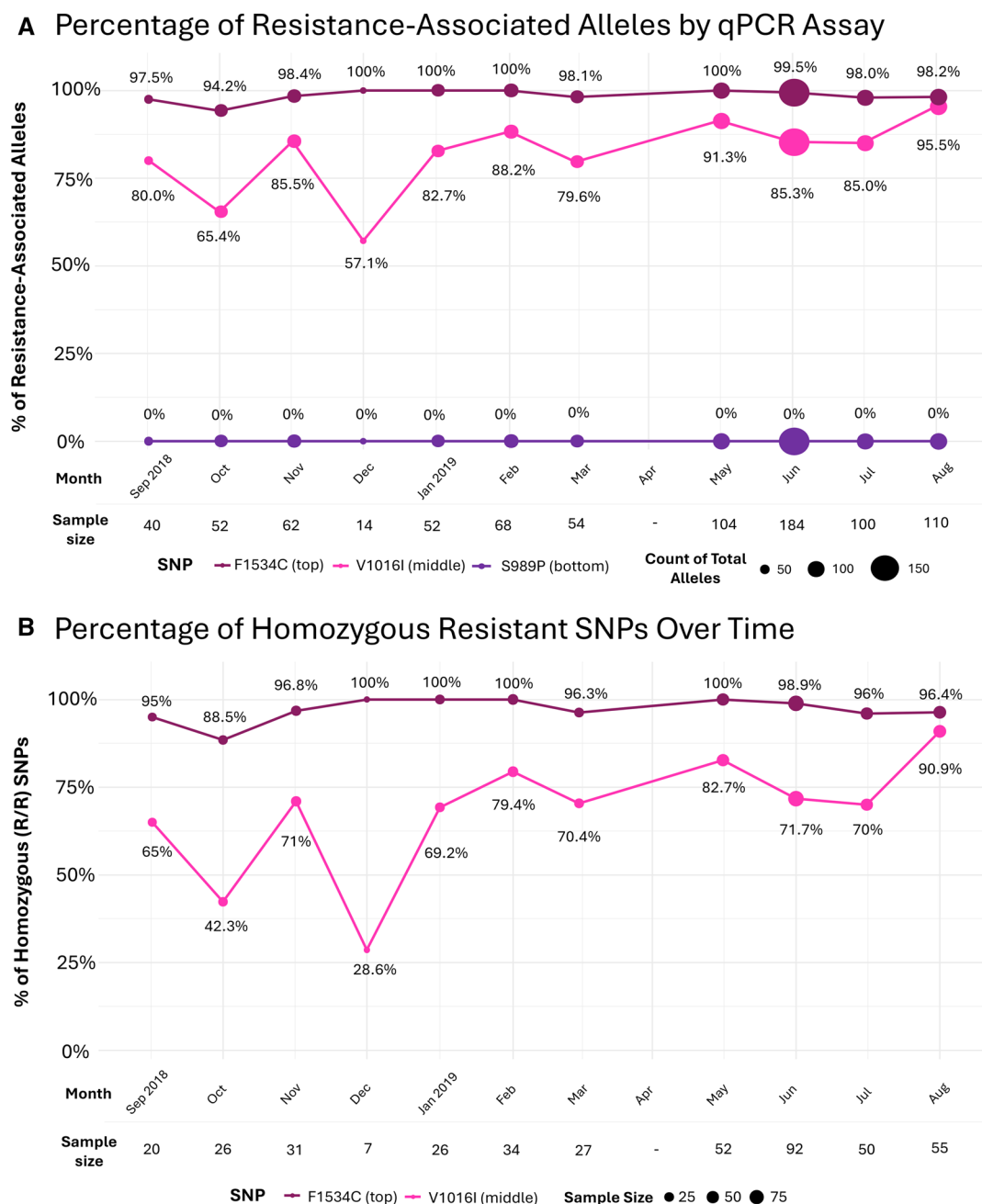


Fig. 3. Percentage of resistant alleles determined by qPCR assay across time (A) and the percentage of alleles homozygous for resistance across time (B). Each colored line indicates a different allele and the percentage of resistance alleles or homozygous alleles during that month. The maroon line (top) represents SNP F1534C, the pink line (middle) indicates SNP V1016I and the magenta line (bottom) is for SNP S989P. The size of each node indicates the count of alleles or mosquitoes tested and is shown numerically as the sample size at the bottom of the figure. Note that SNP S989P is omitted from 3b due to the absence of any homozygous resistance alleles in this study (as shown also in 3A). One sample for the La Salle location had an uncertain collection date and is excluded from temporal analyses.

increase in homozygous resistance genotype frequency was observed for the 1016 locus over the length of the study period. Notably, the cooccurrence of the V1016I and F1534C resistance alleles has been shown in numerous studies to be associated with increased pyrethroid resistance relative to the F1534C mutation alone (Chen et al. 2020). Whereas the V1016I mutation alone has contradictory effects in different populations of mosquitoes and study locations. Current literature suggests that V1016I may be enhancing the effect of or reducing fitness costs conferred by co-occurring mutations (Smith et al. 2016).

This could potentially explain the co-occurrence of the V1016I allele in all mosquitoes with the F1534C allele in our study's population of mosquitoes, although we did not perform functional bioassays prior to genotyping which would be necessary to confirm this.

Previous studies in the Port-au-Prince area denoted limited evidence of pyrethroid resistance markers at the 1016 locus in 2012, although a follow-up study of mosquitoes collected in 2017 found 20% to 86% of samples in the Leogane and Gressier arrondissements of the Ouest department had dilocus *kdr*

mutations at the 1016 and 1534 loci (McAllister et al. 2012, Estep et al. 2024). These findings are noteworthy given all 4 of our study sites report 70–85.7% of *Ae. aegypti* mosquitoes have a dilocus *kdr* mutation, with allelic frequency of resistance SNPs greater than 83.2% and 98.5% for loci 1016 and 1534, respectively. Notably, the study in 2012 did not detect the V1016G mutation and only detected the V1016I in 16% of mosquitoes sampled from Port-au-Prince in May & June 2010 (McAllister et al. 2012). The current study similarly reports only V1016I mutations, with a frequency of 85% in mosquitoes collected in 2018 and 2019. The work by Estep and colleagues provides a useful parallel, where they found both 1016 and 1534 SNPs present in every location they tested. However they describe distinct variability in the resistance allelic frequencies at each site, while we report similar rates of resistance alleles across all 4 sites for all 3 genes examined (Estep et al. 2024).

Pyrethroid-treated bed nets and pyrethroid aerosol sprays are widely available in Haiti and may explain our findings and the steady increase in insecticide resistance across time (Estep et al. 2024). We find this especially relevant given findings in Haiti that permethrin-treated bed nets were shown not effective in preventing malaria (Steinhardt et al. 2017). These findings, along with the ongoing humanitarian crisis in Haiti, all pose challenges which hamper public health efforts regarding the prevention and control of mosquito-borne diseases within the country. These detrimental impacts on public health more broadly emphasize the importance of employing well-tailored and effective disease control methods, especially in the context of constrained resources. We emphasize here the importance of methods beyond broad-spectrum, untailored insecticide applications.

Genotype classifications do not necessarily translate to phenotypic resistance characteristics. Therefore, given the increase in frequency of resistance genotypes, phenotypic resistance studies should be conducted to assess how these genotype changes impact insecticidal activity. Future screening of *Ae. aegypti* mosquitoes should capture other VSSC regions to better characterize cooccurrence of alleles that are additive to resistance and to monitor for the spread of non-native haplotypes to new geographical areas (Auteri et al. 2018, Uemura et al. 2024). Additionally, upregulated detoxification proteins such as glutathione-S-transferases and cytochrome p450s should be included in surveillance screenings due to their implication in pyrethroid resistance as has been demonstrated in other studies (Sene et al. 2021).

Surveillance of mosquitoes for the presence of *kdr* mutations is an important part of effective integrated pest management, as it can inform the need for alternative insecticides to be used. This study reports an increased frequency of *kdr* resistance alleles and genotype markers within *Ae. aegypti* mosquitoes collected from the Ouest department of Haiti, with more than 70% of samples showing evidence of dilocus *kdr* mutations. These findings indicate the need for further studies genetically characterizing other resistance markers and phenotypic studies in assessing the efficacy of pyrethroid insecticides on local *Ae. aegypti* populations.

Author Contributions

Jessy Motes (Data curation [equal], Formal analysis [equal], Writing—original draft [equal], Writing—review & editing [equal]), Bryna Wilson (Investigation [equal], Methodology [equal]), Ian Alexander Pshea-Smith (Formal analysis [equal], Software [equal], Visualization [equal], Writing—original draft

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Supplementary Material

Supplementary material is available at *Journal of Medical Entomology* online.

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Conflicts of Interest

None declared.

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